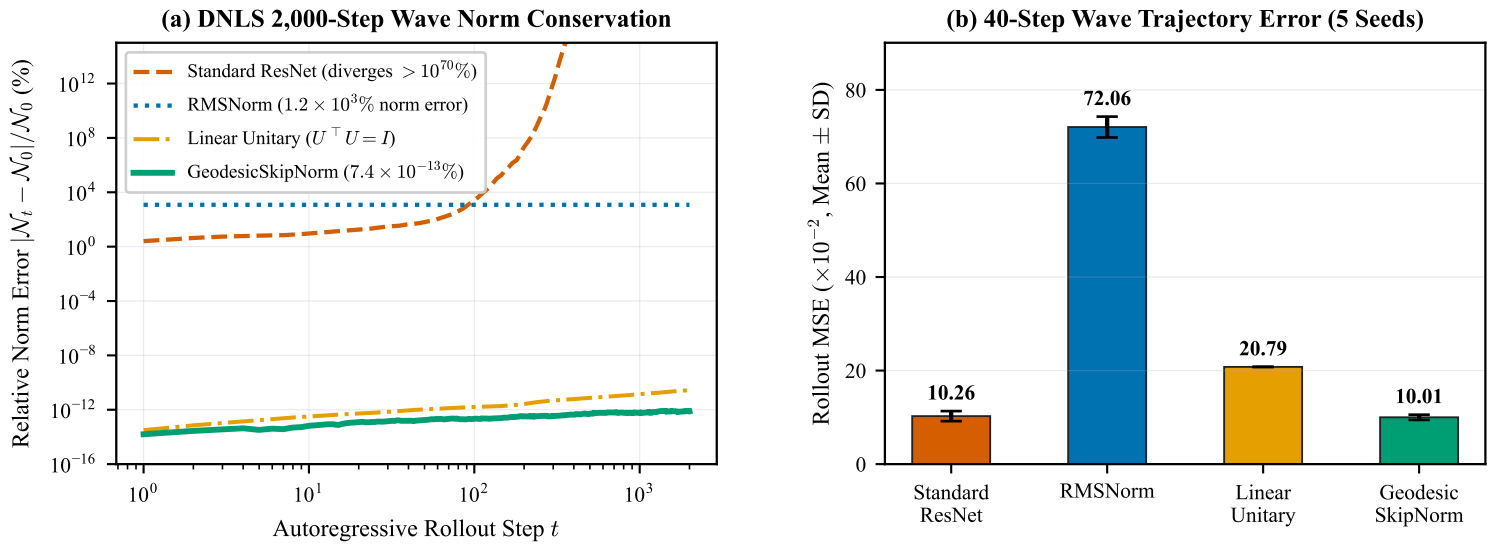
(a) DNLS 2,000-Step Wave Norm Conservation
Standard ResNet (diverges $> 10^{70}\%$)
RMSNorm ($1.2 \times 10^{3}\%$ norm error)
Linear Unitary ($U^\top U = I$)
GeodesicSkipNorm ($7.4 \times 10^{-13}\%$)
Relative Norm Error $|\mathcal{N}_t - \mathcal{N}_0|/\mathcal{N}_0$ (%)
Autoregressive Rollout Step t
(b) 40-Step Wave Trajectory Error (5 Seeds)
Rollout MSE ($\times 10^{-2}$, Mean $\pm$ SD)
72.06
10.26
20.79
10.01
Standard ResNet
RMSNorm
Linear Unitary
Geodesic SkipNorm